Anastasios Tsolakis

Robotics and Control Engineer
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Portfolio
LinkedIn
Scholar
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Profile

Robotics and Control Engineer (PhD, TU Delft) with expertise in dynamics, motion planning, fault-tolerant control, and ROS2-based system integration. Strong mechanical engineering foundation (kinematics, dynamics, mechanisms, design, CAD/CAE) and hands-on deployment of autonomous platforms (ASVs, UGVs). Solid background in dynamical systems, control, estimation, and optimization, with experience translating advanced research into validated engineering solutions in safety-critical environments. Collaborative team player with international, interdisciplinary experience and strong communication skills.

Core Skills & Competencies

- **Robotics software & system integration:** ROS 2 (simulation to deployment), sensor/actuator integration, system validation and testing, model-based systems engineering principles
- **Motion planning & control:** Real-time NMPC, state estimation, trajectory optimization, fault diagnosis, fault-tolerant and safety-critical autonomous systems
- **Programming:** C++ (real-time, mission-level control), Python, Linux, MATLAB/Simulink
- **Engineering tools:** Git; CAD (SolidWorks, Inventor, Creo Parametric); FEA/CAE (ANSA, Altair MotionView)
- **Languages:** Greek (native), English (fluent), French (A1, learning)

Experience

Independent Engineering Projects (listed below)
Jun 2025 - Present

Personal Projects

- Ongoing development of robotics and control software with focus on system integration, testing, and operational readiness.
- Hands-on work on autonomous platforms, sensor integration, real-time control, and validation workflows.

Research & Development (PhD)
Dec 2020 - Jun 2025

TU Delft, The Netherlands

- Designed and validated real-time motion planning and control algorithms enabling safe autonomous navigation under faults, uncertainty, and operational constraints.
- Implemented and integrated ROS-based planning and control pipelines, extensively tested in simulation and prepared for real-world deployment on autonomous platforms.
- Developed automated testing, logging, and performance evaluation workflows for systematic validation and reproducibility.
- Provided technical supervision to MSc students on control design, software architecture, experiments, and validation.

Teaching Assistant
Nov 2019 - Jan 2020

TU Delft, The Netherlands

- Supported MSc-level course on nonlinear systems theory, guiding and evaluating students in assignments.

Mechanical Engineering Intern
Jul 2016 - Oct 2016

Maniatopoulos S.A. IDEAL BIKES, Patras, Greece

- Designed and prototyped components and tools for a cargo e-bike.
- Performed statistical quality analysis to reduce frame defects and improve manufacturing robustness.

Formula Student Team Member
Jul 2013 - Oct 2014

Aristotle Racing Team, Thessaloniki, Greece

- Led design, structural analysis (CAD/CAE), and manufacturing of the vehicle's steel tubular frame.
- Coordinated subsystem integration, subsystem interfaces, assembly sequencing, rule-compliance, and validation.
- Ensured competition readiness; achieved top-10 placements internationally.

Education

PhD in Cognitive Robotics
Dec 2020 – Jun 2025

TU Delft, The Netherlands

- Thesis: *"Rule-compliant and Fault-tolerant Motion Planning: With Application to Autonomous Surface Vehicles"*.
- Developed real-time algorithms for collision avoidance with rule compliance, fault diagnosis, fault reconfiguration, and robustness to uncertainties, resulting in safer and more reliable navigation compared to existing solutions.
- Collaborated within an international, interdisciplinary research team and co-supervised MSc students on projects related to autonomous navigation.

MSc in Systems and Control — *Sep 2018 – Dec 2020*

TU Delft, The Netherlands — GPA: 8.56/10 ***Cum Laude***

- Thesis: *"Distributed IDA-PBC for Nonholonomic Mechanical Systems"*. Designed a distributed passivity-based control algorithm for a fleet of heterogeneous, underactuated, and nonholonomic robots.
- Specialized in control theory, dynamical systems, optimization, and adaptive/predictive/robust control methods.
- Collaborated in small international teams to deliver high-quality results in demanding course projects.

M.Eng. in Mechanical Engineering (Integrated BSc & MSc equivalent) — *Sep 2011 – Jul 2018*

Aristotle University of Thessaloniki, Greece — GPA: 8.08/10

- Thesis: *"The Effect of Power Distribution Architectures and Torque Vectoring in Vehicle Dynamics"*. Investigated vehicle dynamics under different power distribution architectures and developed a torque-vectoring controller.
- Specialized in mechanical design, analysis, synthesis, and control of mechanical systems.
- Member of Formula Student team; collaborated to deliver a competitive racing vehicle for international competitions.

AWARDS & DISTINCTIONS

- MSc Systems & Control *Cum Laude*, TU Delft.
- Young Author Award nominations (IFAC-MICNON 2021, IFAC-CAMS 2022).
- Formula Student: 4th overall (FS Italy); 12th overall (FS Germany); top-10 in Engineering Design (FS Hungary).

PERSONAL PROJECTS

- **nav-mpc — Real-time Nonlinear MPC Framework for Autonomous Navigation** — *Nov 2025 – Present*
  Nonlinear MPC · OSQP · Embedded Control · Python · ROS2
  - Built a real-time NMPC framework with automatic reformulation from symbolic NMPC to LTV-QP reformulation with fast Cython-accelerated evaluation.
  - Real-time QP solution using OSQP, achieving sub-millisecond solution times.
  - Integrated simulation, logging, and benchmarking tools to support systematic validation and reproducible testing.
  - Demonstrated on dynamic systems and mobile robots; designed for embedded use on UGV/UAV/USV platforms.
- **Autonomous Rover — ROS2-Based Mobile Robot with Full Onboard Autonomy** — *Sep 2025 – Present*
  Rover · Raspberry Pi 5 · LiDAR · IMU · Odometry · ROS2
  - Designed and assembled an autonomous rover platform; performed hardware/software bring-up, 3D-printed parts, custom mounting, cable management, logging, and field-run analysis.
  - Implemented EKF-based state estimation from wheel odometry, IMU, and 2D LiDAR; prepared for closed-loop autonomy tests.
  - Modular onboard architecture separating perception/SLAM and planning/control for reliable real-time operation.
- **FLARE — Solar-Powered UAV for Persistent Wildfire Surveillance** — *Sep 2025 – Present*
  UAVs · Solar Power · Environmental Monitoring
  - Developed a wildfire spread simulator using cellular automata and GeoTIFF-based vegetation maps.
  - Conducted payload studies and preliminary airframe sizing and optimization for $> 24\,\mathrm{h}$ endurance.
  - Conceptual design of autonomy stack for a solar-powered autonomous glider with thermal and RGB payloads.
  - Following a model-based systems engineering (MBSE) approach using the V-model, stakeholder analysis, operational scenarios, functional analysis, and requirements definition.
  - Submitted as an MSCA Postdoctoral Fellowship (MSCA-PF) research proposal.

PUBLICATIONS (SELECTED)

- **Tsolakis, A.**, Negenborn, R., Reppa, V., Ferranti, L. *Model Predictive Trajectory Optimization and Control for Autonomous Surface Vessels Considering Traffic Rules*. IEEE Trans. Intell. Transp. Syst., 25(8):9895–9908.
- **Tsolakis, A.**, Ferranti, L., Reppa, V. *Set-Membership Estimation for Fault Diagnosis of Nonlinear Systems*. In *Proc. European Control Conference (ECC)*, Thessaloniki, Greece, 24–27 Jun. 2025.
- **Tsolakis, A.**, Benders, D., De Groot, O., Negenborn, R., Reppa, V., Ferranti, L. *COLREGs-aware Trajectory Optimization for Autonomous Surface Vessels*. IFAC-PapersOnLine 55(31):269–274.
- **Tsolakis, A.**, Keviczky, T. *Distributed IDA-PBC for a Class of Nonholonomic Mechanical Systems*. IFAC-PapersOnLine 54(14):275–280.